

Generation Interconnection System Impact Study Report

For Queue Project Q17-0829

Smithfield 138 kV

October 04, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0829. The Interconnection Customer proposes a 145.5 MW solar photovoltaic facility interconnecting at the Smithfield 138 kV substation in the Virginia area. The request was received on July 18, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 350 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$440,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The LITTLE ROCK / MCPHERSON 2 138.0 kV Line is overloaded to 105.7% of its 234 MVA emergency rating for the loss of the PRINCE / PUERTO NUEVO 1520 138.0 kV line. The project contributes 20.7% DFAX (30.0 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LITTLE ROCK / MCPHERSON 2 138.0 kV	Reconductor line	\$440,000	100% (\$440,000)
	Total allocated to project		\$440,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.